A Geometric Proof of the Leibniz Formula for π

via Gibbs Phasor Chain Convergence

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Abstract

We present a geometric proof of the Leibniz formula $\pi/4 = 1-1/3+1/5-1/7+\cdots$ in which the value π is *constructed*, not imported from any external identity. The proof rests on the phasor chain $\mathcal{C}_N(\theta)$ formed by the partial sums of a Fourier series: a polygonal path in $\mathbb{C}$ whose discrete curvature $\Delta\phi = 2\theta$ is controlled by a single parameter. At the Gibbs angle $\theta = \pi/(N+1)$, the chain folds into a regular $(N+1)$-gon; at $\theta = \pi/2$, it collapses onto the imaginary axis, and its endpoint coordinate *is* the Leibniz partial sum. We show that the endpoint velocity with respect to θ is the odd-harmonic Dirichlet kernel, so the Leibniz sum arises as the integral of this kernel from the aligned state to the folded state. In the limit $N \to \infty$, the polygon becomes a circle, the chord function $m\sin(u/m)$ tends to the arc parametrization u, and the integral evaluates to $\pi/4$. The Leibniz formula is thereby revealed as the one-dimensional projection of a complex phasor polygon that has curled into a circle—the final shadow cast by Archimedes' exhaustion, rebuilt from Fourier harmonics.

1 Introduction

The Leibniz formula for π,

$$\frac{\pi}{4} = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots , \tag{1}$$

is among the most celebrated identities in mathematics. Discovered by Mādhava of Saṅgamagrāma circa 1350 [1, 2], and independently by Gregory (1671) and Leibniz (1676), it is conventionally proved by evaluating the Maclaurin series of $\arctan(x)$ at $x = 1$.

All standard proofs share a common feature: the value π enters the argument *externally*—through the known identity $\arctan(1) = \pi/4$, the periodicity of $e^{i\theta}$, or the normalization of the Gaussian integral. In each case, π is *imported*, not *derived*.

In this paper we take a different approach. We treat the partial sums of a Fourier series as a phasor chain in the complex plane—a polygonal path whose shape is controlled by a single curvature parameter θ. As θ is varied from 0 to $\pi/2$, the chain deforms continuously from a straight line through a regular polygon to a fully folded alternating sum. The Gibbs polygon and the Leibniz series are thereby revealed as two states of the *same geometric object*, connected by a smooth deformation.

Building on the result established in [3] that the Gibbs phasor chain at N harmonics folds into a regular $(N+1)$-gon converging to an apeirogon (a circle), we show that:

(i) The parameter θ acts as a discrete curvature: $\Delta\phi = 2\theta$ is the turning angle between consecutive arms, deforming the chain from aligned ($\theta = 0$) through circular ($\theta = \pi/(N{+}1)$) to folded ($\theta = \pi/2$).

(ii) The endpoint velocity $dP_{N+1}/d\theta$ is the odd-harmonic Dirichlet kernel, so the Leibniz partial sum arises as the integral of this kernel.

(iii) In the limit $N \to \infty$, the polygon-to-circle transition replaces the chord function $m\sin(u/m)$ with the arc parametrization u, and the integral evaluates to $\pi/4$.

(iv) The convergence rate decomposes into a polygon-to-circle bulk error of $O(1/N^2)$ and a Gibbs tail residual of $O(1/N)$.

The value π is not assumed—it is the ratio that the phasor geometry forces into existence as the polygon curls into a circle.

2 The Phasor Chain

2.1 Fourier partial sums as vector chains

Consider the Fourier series of the square wave on $[-\pi, \pi]$:

$$f(x) = \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{\sin\big((2k+1)x\big)}{2k+1} \,. \tag{2}$$

The N-th partial sum evaluated at $x = \theta$ can be written as the imaginary part of a sum of complex exponentials:

$$S_N(\theta) = \mathrm{Im}\left(\frac{4}{\pi} \sum_{k=0}^{N} \frac{e^{i(2k+1)\theta}}{2k+1} \right). \tag{3}$$

Each term

$$\mathbf{v}_k(\theta) = \frac{1}{2k+1}\, e^{i(2k+1)\theta} \tag{4}$$

is a *phasor arm*: a vector in $\mathbb{C}$ of length $|\mathbf{v}_k| = 1/(2k+1)$ and phase angle $(2k+1)\theta$.

Definition 1 (Phasor chain)**.** *The* phasor chain $\mathcal{C}_N(\theta)$ *is the polygonal path formed by placing the arm vectors* $\mathbf{v}_0, \mathbf{v}_1, \ldots, \mathbf{v}_N$ *tip-to-tail from the origin:*

$$P_0 = 0, \qquad P_{j+1} = P_j + \mathbf{v}_j, \qquad j = 0, 1, \ldots, N.$$

The endpoint $P_{N+1}(\theta)$ *gives the complex partial sum.*

2.2 The $(N{+}1)$-gon theorem (established result)

At the Gibbs overshoot angle $\theta_N^* = \pi/(N{+}1)$, the angular separation between consecutive arms becomes uniform:

$$\arg(\mathbf{v}_{k+1}) - \arg(\mathbf{v}_k) = \big(2(k{+}1) + 1\big)\theta_N^* - (2k+1)\theta_N^* = 2\theta_N^* = \frac{2\pi}{N+1} \,. \tag{5}$$

This is the central angle of a regular $(N{+}1)$-gon. Since the angular increments are equal, the arm vectors are chords of a common circle, and the phasor chain inscribes a regular $(N{+}1)$-gon. As $N \to \infty$, the polygon converges to an apeirogon—a circle [3].

Remark 2. *The angular separation* (5) *is* exact *for all finite N; the polygon regularity is an algebraic identity, not an approximation.*

3 The Curvature Deformation

The parameter θ controls the shape of the phasor chain through a single geometric quantity: the *discrete curvature*

$$\Delta\phi(\theta) = \arg(\mathbf{v}_{k+1}) - \arg(\mathbf{v}_k) = 2\theta\,. \tag{6}$$

This turning angle between consecutive arms is independent of k—every joint in the chain bends by the same amount. As θ varies from 0 to $\pi/2$, the chain passes through three characteristic regimes:

I. Aligned state ($\theta = 0$, $\Delta\phi = 0$). All arms point in the same direction along the positive real axis. The chain is a straight segment of total length

$$L_N = \sum_{k=0}^{N} \frac{1}{2k+1}\,, \tag{7}$$

and the endpoint is $P_{N+1}(0) = L_N \in \mathbb{R}$. The curvature is zero.

II. Polygonal state ($\theta = \pi/(N+1)$, $\Delta\phi = 2\pi/(N+1)$). The uniform turning angle forces the arms to become chords of a circle, forming a regular $(N+1)$-gon. The total angular span of the arc is

$$\alpha_N = N \cdot \frac{2\pi}{N+1} = \frac{2N\pi}{N+1}\,, \tag{8}$$

which approaches 2π as $N \to \infty$. Each arm $\mathbf{v}_k$, viewed as a chord subtending angle $\Delta\phi$ on a local circle, determines a local radius

$$r_k = \frac{|\mathbf{v}_k|}{2\sin(\pi/(N+1))} = \frac{1}{(2k+1)\cdot 2\sin(\pi/(N+1))}\,. \tag{9}$$

The chord-to-arc ratio for each arm is

$$\frac{|\mathbf{v}_k|}{r_k\,\Delta\phi} = \frac{2\sin\bigl(\pi/(N+1)\bigr)}{2\pi/(N+1)} = \operatorname{sinc}\left(\frac{1}{N+1}\right), \tag{10}$$

which is *independent of* k. This k-independence is the critical structural fact: every arm, regardless of its length, converts between chord and arc at the same ratio.

III. Folded state ($\theta = \pi/2$, $\Delta\phi = \pi$). Each arm is rotated 180° relative to its predecessor. The chain folds back on itself along a single axis. The arms evaluate to

$$\mathbf{v}_k(\pi/2) = \frac{e^{i(2k+1)\pi/2}}{2k+1} = \frac{i^{2k+1}}{2k+1} = \frac{i\,(-1)^k}{2k+1}\,, \tag{11}$$

so every arm points along the imaginary axis, alternating sign. The endpoint is

$$P_{N+1}(\pi/2) = i\sum_{k=0}^{N} \frac{(-1)^k}{2k+1} = i\,S_N^{\mathrm{Leib}}\,. \tag{12}$$

The imaginary coordinate of the endpoint *is* the Leibniz partial sum.

Remark 3. *The three states—aligned, polygonal, folded—are not separate constructions. They are the same phasor chain at different curvature settings. The Gibbs polygon and the Leibniz series are two views of one geometric object.*

4 The Endpoint Velocity

To connect the polygonal state (where the circle geometry lives) to the folded state (where the Leibniz sum lives), we study how the endpoint moves as θ changes.

Proposition 4 (Endpoint velocity)**.** *The velocity of the phasor chain endpoint with respect to the curvature parameter θ is the odd-harmonic Dirichlet kernel:*

$$\frac{d}{d\theta} P_{N+1}(\theta) = i \sum_{k=0}^{N} e^{i(2k+1)\theta} = i\, D_N^{\mathrm{odd}}(\theta)\,, \tag{13}$$

where $D_N^{\mathrm{odd}}(\theta)$ denotes the odd-harmonic Dirichlet kernel.

Proof. Differentiating the endpoint (3) term by term:

$$\frac{d}{d\theta} P_{N+1}(\theta) = \sum_{k=0}^{N} \frac{i(2k+1)}{2k+1}\, e^{i(2k+1)\theta} = i \sum_{k=0}^{N} e^{i(2k+1)\theta}\,. \qquad \square$$

The odd-harmonic Dirichlet kernel has the well-known closed form (see, e.g., [4], Ch. II):

$$D_N^{\mathrm{odd}}(\theta) = \sum_{k=0}^{N} e^{i(2k+1)\theta} = \frac{e^{i\theta}\big(1 - e^{i2(N+1)\theta}\big)}{1 - e^{i2\theta}} = e^{i(N+1)\theta}\, \frac{\sin\big((N+1)\theta\big)}{\sin\theta}\,. \tag{14}$$

Remark 5. *The "speed" $|dP_{N+1}/d\theta|$ at which the endpoint traverses the complex plane is governed by the magnitude of the Dirichlet kernel. This magnitude exhibits the same oscillatory peaks that produce the Gibbs overshoot. The phasor chain's deformation and the Gibbs phenomenon are driven by the same engine.*

The endpoint position at any θ is recovered by integrating the velocity from the aligned state:

$$P_{N+1}(\theta) = P_{N+1}(0) + i \int_0^\theta D_N^{\mathrm{odd}}(t)\, dt = L_N + i \int_0^\theta \frac{\sin\big((N+1)t\big)}{\sin t}\, e^{i(N+1)t}\, dt\,. \tag{15}$$

For the imaginary part of $P_{N+1}(\theta)$ (which gives the Fourier partial sum), we extract:

$$\mathrm{Im}\big[P_{N+1}(\theta)\big] = \int_0^\theta \mathrm{Re}\big[D_N^{\mathrm{odd}}(t)\big]\, dt = \int_0^\theta \frac{\sin\big((N+1)t\big)\cos\big((N+1)t\big)}{\sin t}\, dt\,. \tag{16}$$

Using the identity $\sin\alpha\cos\alpha = \frac{1}{2}\sin 2\alpha$ with $\alpha = (N+1)t$:

$$\mathrm{Im}\big[P_{N+1}(\theta)\big] = \frac{1}{2}\int_0^\theta \frac{\sin\big(2(N+1)t\big)}{\sin t}\, dt\,. \tag{17}$$

This integral representation—which in the classical literature is derived from summation identities [4]—here arises *naturally* as the integral of the endpoint velocity. The Dirichlet kernel is not an external tool; it is the intrinsic speed of the phasor chain's deformation.

5 The Proof

Theorem 6 (Geometric Leibniz formula)**.**

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} = \frac{\pi}{4}\,.$$

Proof. **Step 1: The Leibniz sum as a phasor projection.** From (12), the Leibniz partial sum is the imaginary coordinate of the phasor chain endpoint at $\theta = \pi/2$:

$$S_N^{\text{Leib}} = \sum_{k=0}^{N} \frac{(-1)^k}{2k+1} = \operatorname{Im}\big[P_{N+1}(\pi/2)\big]/1\,, \tag{18}$$

since $P_{N+1}(\pi/2) = i\,S_N^{\text{Leib}}$ lies entirely on the imaginary axis. The Leibniz sum is the displacement of the fully folded phasor chain.

Step 2: The velocity integral. From Proposition 4 and equation (17), the Leibniz partial sum equals the integral of the endpoint velocity from the aligned state ($\theta = 0$) to the folded state ($\theta = \pi/2$):

$$S_N^{\text{Leib}} = \frac{1}{2}\int_0^{\pi/2} \frac{\sin\big((2N{+}2)\,t\big)}{\sin t}\,dt\,. \tag{19}$$

This integral has a direct geometric meaning: it is the cumulative displacement of the chain endpoint along the imaginary axis as the curvature $\Delta\phi = 2\theta$ is increased from 0 (straight) to π (fully folded), driven at each instant by the Dirichlet kernel.

Step 3: The polygon-to-circle limit. Set $m = 2N + 2$ and substitute $u = mt$, so that $dt = du/m$ and $\sin t = \sin(u/m)$:

$$S_N^{\text{Leib}} = \frac{1}{2}\int_0^{m\pi/2} \frac{\sin u}{m\sin(u/m)}\,du\,. \tag{20}$$

The denominator $m\sin(u/m)$ is the *chord function*: it is the chord length of an arc of angular measure u/m on a circle of m sides, scaled to unit radius. This factor converts the smooth arc parametrization into the discrete chord parametrization of the polygon.

As $m \to \infty$ (i.e., $N \to \infty$), the polygon becomes a circle. The chord function converges to the arc parametrization:

$$m\sin\left(\frac{u}{m}\right) = u - \frac{u^3}{6m^2} + O\left(\frac{u^5}{m^4}\right) \;\longrightarrow\; u\,. \tag{21}$$

This is the geometric heart of the proof. The transition $m\sin(u/m) \to u$ is precisely the statement that chords become arcs as the number of polygon sides tends to infinity —the same principle underlying Archimedes' method of exhaustion [6].

Step 4: Evaluation of the limit. By dominated convergence (the integrand is bounded by $1/\sin(u/m) \le m/u$ for $u > 0$, and the tail $\int_{m\pi/2}^{\infty} \sin(u)/u\,du \to 0$):

$$\lim_{N\to\infty} S_N^{\text{Leib}} = \frac{1}{2}\int_0^{\infty} \frac{\sin u}{u}\,du = \frac{1}{2}\cdot\frac{\pi}{2} = \frac{\pi}{4}\,. \tag{22}$$

The Dirichlet integral $\int_0^\infty \sin(u)/u\, du = \pi/2$ is itself a statement about circular geometry: it is the total signed projection accumulated by a unit-frequency phasor as it sweeps from 0 to ∞—a half-rotation of the limiting circle. The value $\pi/2$ is not imported from an external source; it is the content of the arc-length definition of π applied to the circle that the polygon has become.

Step 5: Conclusion. Therefore:

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} = \frac{\pi}{4}. \qquad \square$$

Remark 7 (The role of π)**.** *In Steps 1–3, the symbol π appears only as an integration limit ($\theta = \pi/2$) and in the polygon parameters ($\theta_N^* = \pi/(N+1)$). The* numerical value $\pi = 3.14159\ldots$ *enters solely in Step 4, through the Dirichlet integral, where it is* produced *by the polygon-to-circle limit—not assumed.*

6 The Convergence Rate

The curvature deformation framework provides a geometric decomposition of the convergence error into two sources with distinct rates, answering a natural question: if the polygon-to-circle geometry is so clean, why does the Leibniz series converge so slowly?

6.1 The geometry of the complex chord

Each phasor arm $\mathbf{v}_k$ acts as a chord in the circumscribed circle. The fundamental identity established in (10) is that the chord-to-arc ratio is uniform across the entire chain for a given N:

$$\frac{|\mathbf{v}_k|}{\ell_k} = \operatorname{sinc}\left(\frac{1}{N+1}\right), \tag{23}$$

where $\ell_k = r_k\, \Delta\phi$ is the arc segment corresponding to the k-th chord. Expanding the sinc function:

$$\operatorname{sinc}(x) = \frac{\sin(\pi x)}{\pi x} = 1 - \frac{\pi^2 x^2}{6} + O(x^4). \tag{24}$$

Substituting $x = 1/(N+1)$, the total chord length (the phasor chain length L_N) and the total arc length $\mathcal{A}_N$ are related by:

$$L_N = \mathcal{A}_N \left(1 - \frac{\pi^2}{6(N+1)^2} + O\left(\frac{1}{N^4}\right)\right). \tag{25}$$

The "jaggedness" of the polygon—the discrepancy between the chord path and the smooth arc—vanishes as $O(1/N^2)$. The π-generating engine, the polygon folding into a circle, is *efficient.*

6.2 Error decomposition

From (20), the total error at finite N (with $m = 2N+2$) decomposes as:

$$S_N^{\text{Leib}} - \frac{\pi}{4} = \underbrace{\frac{1}{2}\int_0^{m\pi/2} \sin u \left(\frac{1}{m\sin(u/m)} - \frac{1}{u}\right) du}_{E_{\text{bulk}}:\ \text{polygon-to-circle error}} - \underbrace{\frac{1}{2}\int_{m\pi/2}^{\infty} \frac{\sin u}{u}\, du}_{E_{\text{tail}}:\ \text{Gibbs tail}}. \tag{26}$$

The bulk error E_{bulk}: $O(1/N^2)$. From the chord function expansion (21):

$$\frac{1}{m\sin(u/m)} - \frac{1}{u} = \frac{u}{6m^2} \cdot \frac{1}{1 - u^2/(6m^2) + \cdots} \approx \frac{u}{6m^2} \quad \text{for } u \ll m\,.$$

The bulk integral is therefore:

$$E_{\text{bulk}} = \frac{1}{2}\int_0^{m\pi/2} \frac{u\sin u}{6m^2}\,du + O(m^{-4}) = O\left(\frac{1}{m^2}\right) = O\left(\frac{1}{N^2}\right).$$

This is the cost of approximating arcs with chords. It is the same error that appears in Archimedes' polygon estimates, and it vanishes quadratically.

The tail error E_{tail}: $O(1/N)$. The tail of the Dirichlet integral represents the displacement that the phasor chain has not yet accumulated at finite N—the harmonics that have been truncated. By integration by parts:

$$\int_{m\pi/2}^{\infty} \frac{\sin u}{u}\,du = \frac{-\cos(m\pi/2)}{m\pi/2} + \int_{m\pi/2}^{\infty} \frac{\cos u}{u^2}\,du = O\left(\frac{1}{m}\right) = O\left(\frac{1}{N}\right).$$

This is the dominant error term. It is the analytic manifestation of the Gibbs overshoot: the "leftover" folding that the finite chain cannot complete.

6.3 Why the Leibniz series is slow

The decomposition (26) reveals a precise geometric answer to a classical question.

The π-generating mechanism—the polygon curling into a circle—converges at rate $O(1/N^2)$. If the error were due solely to the polygon's approximation of the circle, the Leibniz series would converge quadratically.

But the total error is $O(1/N)$, dominated by E_{tail}. The slow convergence is not a deficiency of the geometry; it is a *boundary condition* error. At finite N, the phasor chain has $N+1$ arms. The circular arc they trace is already an excellent approximation to a true circle. What is missing is not curvature precision but *arc length*: the chain is not yet long enough to complete its sweep. The remaining displacement leaks out as the tail of the sinc integral, decaying only as $O(1/N)$.

In the language of the curvature deformation: the engine is fast, the truncation is slow.

7 Discussion

7.1 What is new

The identity (1) is not new. What is new is the *mechanism of proof* and the conceptual framework surrounding it.

In our argument:

(a) π is not invoked from arctan, from Euler's formula, or from any prior series. It arises as the limiting ratio of arc length to radius in a polygon that converges to a circle.

(b) The polygon is not imposed by construction (as in Archimedes' method of exhaustion); it *emerges naturally* from the equal angular spacing of the Fourier phasor chain at the Gibbs overshoot angle.

(c) The Gibbs polygon and the Leibniz series are identified as two states of a single geometric object —the phasor chain $\mathcal{C}_N(\theta)$—connected by a continuous deformation parametrized by the discrete curvature $\Delta\phi = 2\theta$.

(d) The Dirichlet kernel is revealed as the *intrinsic velocity* of this deformation, and the integral representation of the Leibniz sum arises from integrating this velocity—not from external summation formulas.

(e) The sinc function $\sin(u)/u$ is identified as the *chord density* of the limiting circle: the ratio of chord travel to arc travel as the polygon becomes smooth.

(f) The convergence rate of the Leibniz series acquires a geometric decomposition: a polygon-to-circle bulk correction of $O(1/N^2)$ dominated by a Gibbs tail residual of $O(1/N)$.

7.2 Relation to Archimedes

Archimedes computed π by inscribing and circumscribing regular polygons in a circle and bounding the circumference [6]. Our construction inverts this: the polygon arises from the Fourier partial sums, the circle emerges in the limit, and π is recovered from the ratio.

The parallel is precise. Archimedes' polygon has n sides of equal length; our phasor chain has $N+1$ arms of *unequal* length but *equal angular spacing*. In both cases, the polygon-to-circle transition is governed by $n\sin(\pi/n) \to \pi$, which is the content of equation (21). The Gibbs phenomenon *is* Archimedes' exhaustion, performed automatically by the Fourier machine.

7.3 Relation to the Gibbs phenomenon

The Gibbs overshoot factor $\mathrm{Si}(\pi)/(\pi/2) \approx 1.17898\ldots$ is conventionally derived analytically [5]. Our framework reveals it as a geometric ratio: the overshoot is the excess arc length of the first lobe of the sinc function, measured against the half-circle it approximates. The factor $2/\pi$ that converts Gibbs overshoot to the steady-state value is the same π that appears in the Leibniz formula—they are the same geometric constant, seen from two different curvature settings of the phasor chain.

7.4 The Leibniz formula as a shadow

The proof reveals a unifying geometric picture. The Leibniz series is not a sum of fractions that happens to equal $\pi/4$. It is the one-dimensional projection—the *shadow*—of a high-dimensional complex rotation. The $N+1$ phasor arms rotate in $\mathbb{C}$, their chain curling into a polygon that tends to a circle. The alternating partial sum $1 - 1/3 + 1/5 - \cdots$ is the real trace of this complex curling, recorded on the imaginary axis at the moment the curvature reaches its maximum. The value $\pi/4$ is the specific displacement where the infinite folding of the complex arms perfectly balances the circular geometry they have created.

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